

ASHISH VERMA

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EDUCATION

Indian Institute of Technology Guwahati (IITG)
Master of Technology in Computer Science and Engineering

2024 - 2026

Kamla Nehru Institute of Technology Sultanpur (KNIT)
Bachelor of Engineering in Computer Engineering

2018 - 2022

PROJECTS

Design a PEM Algorithm for the LCS Problem | MTech Thesis Project

July 2025 – Present

- **Designing** Parallel Algorithms for the **Longest Common Subsequence (LCS)** problem using **Parallel External Memory (PEM)**, aiming to speed up processing and reduce I/O overhead for very large data.

Logic Synthesis Using Heuristics | 

Feb. 2025 – Apr. 2025

- **Designed** a neural network combining **GCN and embedders** to predict and optimize circuit delay in different scaled AIGs, cutting logic synthesis **time from 3 hours (manual) to 20 minutes** with **$\pm 4.02\%$ error**.
- **Tech stack:** Python, PyTorch, Shell Scripts, ABC tool, Few-Shot Learning, Simulated Annealing, Monte Carlo Optimization.

Multi-Agent PDF Chatbot (RAG) | 

Month Year – Month Year

- **Engineered** a **Retrieval-Augmented Generation (RAG)** system enabling interactive, persistent, and multi-agent conversations with PDF documents, featuring source-cited responses for verifiable and reliable AI interactions.
- **Implemented** a scalable architecture using **ChromaDB** for persistent vector storage and **LangChain** for efficient text chunking and retrieval, allowing seamless agent creation and context management.
- **Tech Stack:** Streamlit, Google Gemini 1.5 Flash, ChromaDB, LangChain, PyMuPDF, Sentence-Transformers.

Blog-Bar - Full-Stack Blog Platform | 

Sept. 2025

- A **Full-stack** blogging platform designed for creating, sharing, and managing articles. It provides a secure environment for authors to publish content and a **dynamic, searchable** experience for readers.
- Engineered a secure **Node.js** API for content management and designed a responsive **React** front-end with features like a rich text editor and an admin dashboard.
- **Tech Stack:** React, Redux, Node.js, Express.js, MongoDB, JWT, and Tailwind CSS.

Secure Hybrid TCP/UDP Client-Server Framework | 

Apr. 2025 – May 2025

- **Engineered** a **hybrid TCP/UDP C++ architecture** for high-speed data transfer, leveraging TCP for reliable session management and UDP for transmission.
- **Implemented** custom **FCFS and Round Robin scheduling policies** alongside a 16-byte symmetric XOR encryption scheme with Base64 encoding.
- **Built** integrated **throughput measurement tools** and debug utilities for performance analysis;
- **Tech Stack:** C++, Socket Programming, TCP/UDP, Encryption.

ACHIEVEMENTS

- **AIR 733**, GATE CSE (**99.40 %ile**) among **123,967 students**
- **Competitive Programming**, Codeforces **Specialist** with a contest rating of **1411**
- **Coding**, Solved **1000+** **Leetcode** Problems honing Data Structures and Algorithms(DSA)

SKILLS

- **Languages:** C/C++, Python
- **Web Technologies:** HTML, CSS, JavaScript, React, Node.js, Express.js
- **Databases:** MongoDB, MySQL
- **Tools:** Git, GitHub, Linux (Elementary), Google Colab, Visual Studio, Windows

RELEVANT COURSEWORK

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|----------------------------------|------------------------------|------------------------------|
| • Design & Analysis of Algorithm | • Introduction to ML (NPTEL) | • High Performance Computing |
| • FM for System Verification | • ML for EDA | • Cloud Computing (NPTEL) |
| • Speech Processing | • Data Structures Lab | • Systems Lab |